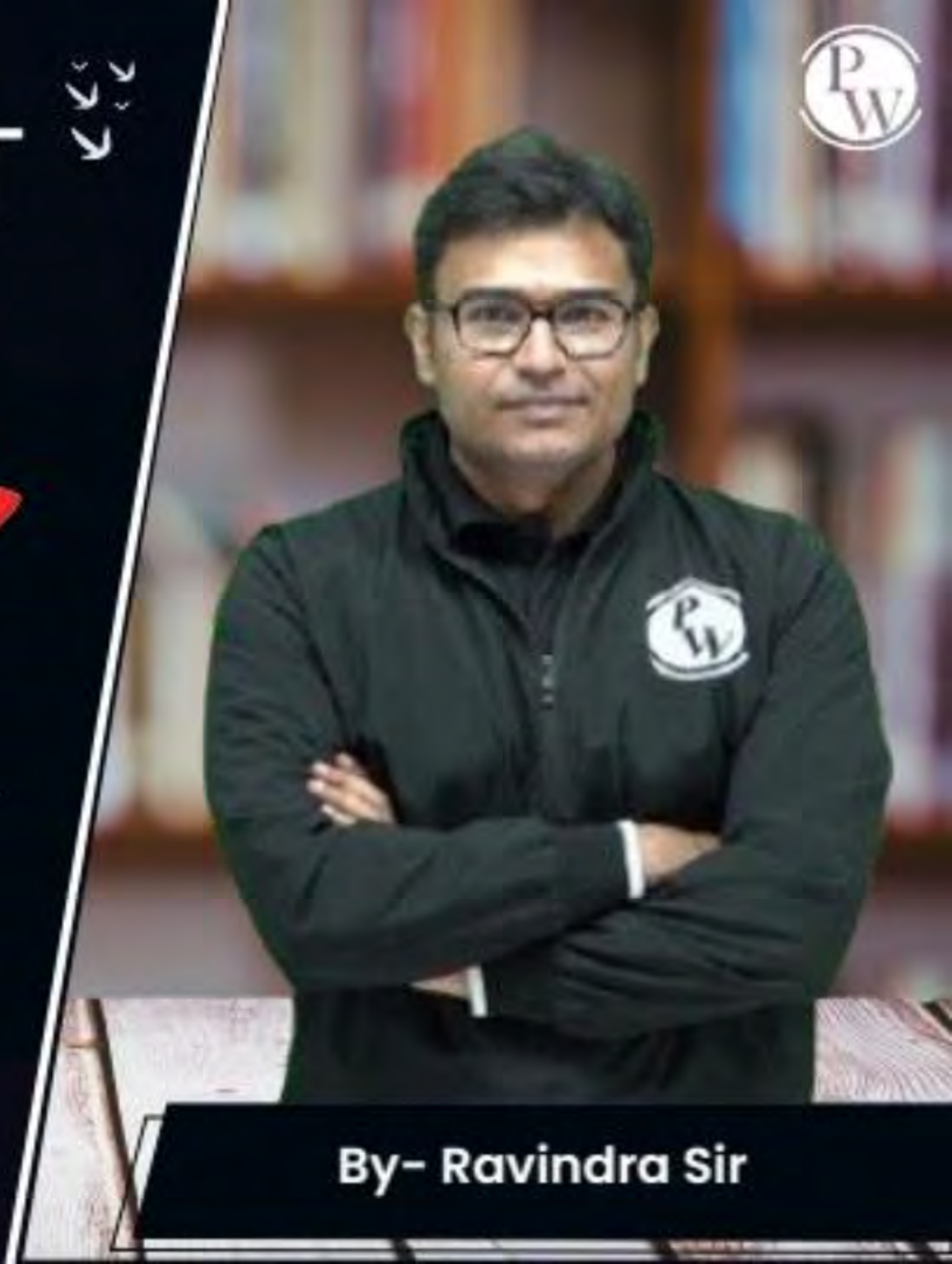
Computer Science & IT

Theory of Computation

Context Free Languages

Lecture No. 2



Recap of Previous Lecture

Topic CFL

Topic

Topic

$$\underbrace{a^n b^m}_{\text{Push}} \underline{c}^{n+m}$$

Topics to be Covered

Topic CFL

Topic

Topic

Limitations of DPDA :-

→ when to push when to pop is not clear, DPDA fails

Even length palindrome → CFL ✗ DCFL

Odd " " → CFL ✗ DCFL

all palindromes → CFL ✗ DCFL

wcw^R → DCFL ✓ CFL ✓

→ How much to push is not clear, then DPDA fails

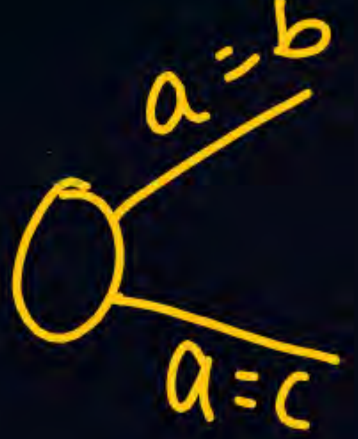
Ex: $\{a^n b^n\} \cup \{a^n b^{2n}\}$

$w / n_a = n_b$ & $n_a = 2n_b$

$\left.\begin{matrix} \\ \\ \end{matrix}\right\}$ DPDA ✗ NPDA ✓

→ Double Comparison

$a^m b^n c^p / m=n \ \& \ m=p$ ✗DPDA ✓NPDA



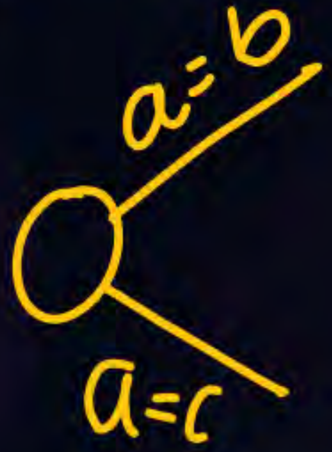
$a^n b^n c^n$ → DPDA✗ NPDA✗ CSL✓

$a^m b^n \,/\, n \le m \le 2n$

CFL but not DCFL

$\Sigma = \{a, b, c\}$

$L = w \,/\, n_a = n_b$ or $n_a = n_c$ NPDA ✓ DPDA ✗



$\underset{\text{Push}}{\underline{a^{m+n}}}\ \underset{\text{Pop}}{\underline{b^n}}\ \underset{\text{pop}}{\underline{c^m}} \,/\, n, m \geq 1$ DCFL ✓ Reg ✗ CFL ✓

$a^m\, b^{m+n}\, c^n$

$\underset{\text{P}}{\underline{a^m}}\ \underset{\text{Pop}}{\underline{b^m}}\ \underset{\text{pu}}{\underline{b^n}}\ \underset{\text{pop}}{\underline{c^n}} \rightarrow$ DCFL ✓ CFL ✓ Reg ✗

$\underset{\text{Pus}}{\underline{a^m}}\ \underset{\text{Pus}}{\underline{b^n}}\ \underset{\text{Pop}}{\underline{c}}{}^{m+n} \,/\, m,n \geq 1$

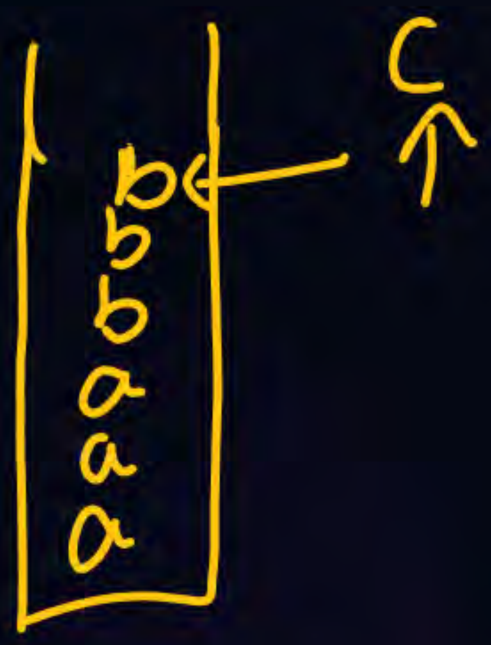
✓ DCFL

$\underset{Pu}{\underline{a}}^m \underset{Po}{\underline{b}}^m \underset{Pu}{\underline{c}}^n \underset{Po}{\underline{d}}^n / n, m \geq 1$

DCFL ✓

$$a^m b^n c^m d^n \,/\, n, m \geq 1$$

CFL × DCFL × CSL ✓



$$\underset{PU}{\underline{a}}^{m}\ \underset{PU}{\underline{b}}^{n}\ \underset{Pop}{\underline{c}}^{n}\ \underset{Pop}{\underline{d}}^{m}$$

DCFL ✓

$$a^m a^i c^m d^k \,/\, m, i, k \geq 1$$

$$a^{\underline{\underline{m+i}}} c^{\underline{\underline{m}}} d^k$$

$$\underline{a^x c}^y d^z \,/\, x > y \,/\, x, y, z \geq 1$$

✓DCFL CFL✓

$a^m b^n / m < n$

✓ DCFL

$(b, \geq 0/$

$a^n b^{2n} / n \geq 1$

✓ DCFL

$a^n \underline{\underline{b}}^{n^2} / n \geq 1$

✓ CSL

✗ CFL

$a^n b^{2n}$

✓ CSL

✗ CFL

$\{ww^R \mid w \in \{a,b\}^*\}$

✗ DCFL

✓ CFL

$wcw^R / w \in (a,b)^*$

$\uparrow$

✓DCFL

$ww \,/\, w \in (a,b)^*$

$\underbrace{abb}_{w}\,\underbrace{abb}_{w}$

CFL ✗

CSL ✓

$a^n b^n c^m / n > m$

$\times$ CFL

$\checkmark$ CSL

$a^n b^n c^n d^n \,/\, n \leq 10^{10}$

$\downarrow$

finite

$\downarrow$

Regular

$a^n b^{2n} c^{3n} / n \geq 1$

CFL X

CSL

$\underline{a^n}\ \underline{b^{2n}}\ \underline{c^m} / n, m \geq 1$

Paa Pop No action

✓ DCFL ✓ CFL

× Reg

$a^n b^{3m} c^{2n} / n, m \geq 1$

✓DCFL, ✗Reg

$x c y \,/\, x, y \in (a,b)^*$

$(a+b)^* c (a+b)^* \to$ Reg

$x c x \,/\, x \in (a,b)^*$ CSL

$\underset{x}{\underline{abb}}\, c\, \underset{x}{\underline{abb}}$

WW^R / WF $(a,b)^*$ $|x| = 10$

↓

finite

↓

Regular

$\underline{ww}w^R / w \in (a, b)^*$

✓ CSL

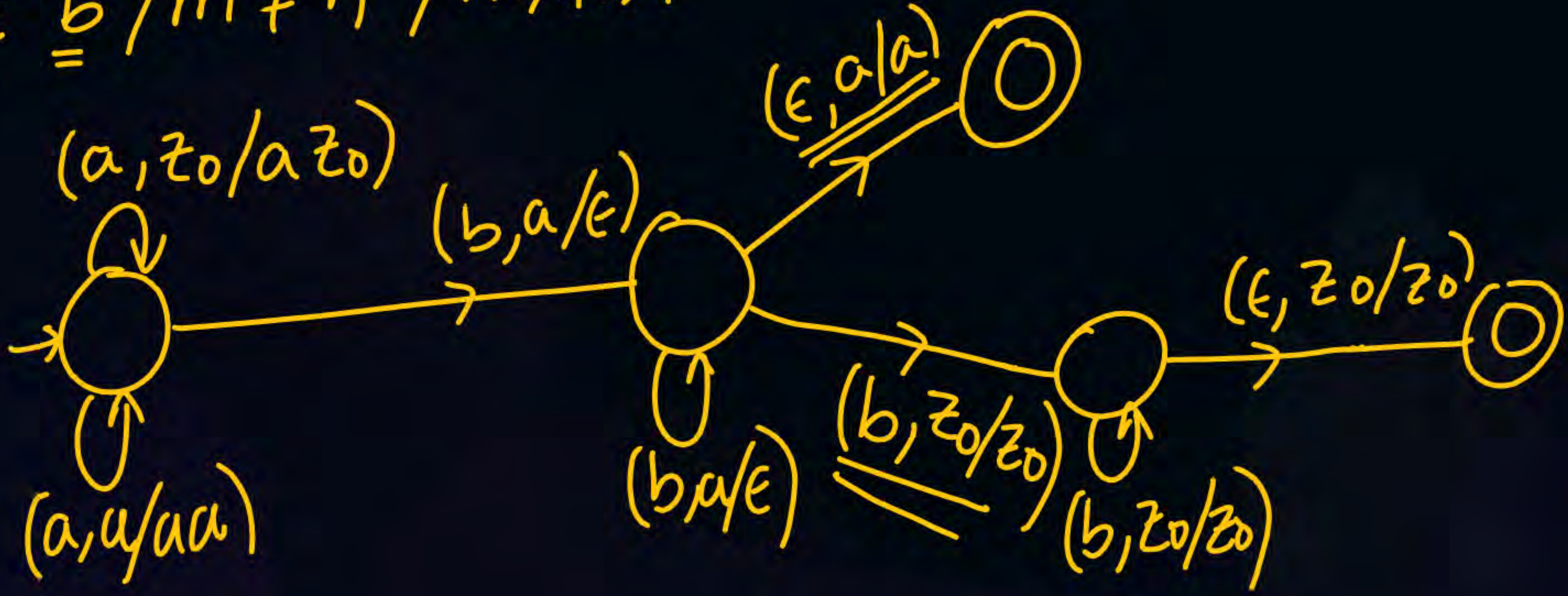
✗ CFL

$a^n b^{3n} / n \geq 1$

✓ CSL

× CFL

$a^m b^n / m \neq n / m, n \geq 1$



$a^i b^j / i \neq 2j+1$

✓ DCFL ✓ CFL

✗ Reg

$a^n b^m / n = 2m+1$

✓DCFL

$a^{n^2} / n \geq 1$	$a^{2^n} / n \geq 1$	$a^{n!} / n \geq 1$	a^m / m is prime
✗ CFL	✗ CFL	✗ CFL	✗ CFL
✓ CSL	✓ CSL	✓ CSL	✓ CSL

a^k / k is even $\rightarrow$ reg

$\underbrace{a^i b^j}\underbrace{c^k} / i > j > k$

✓ CSL

2 min

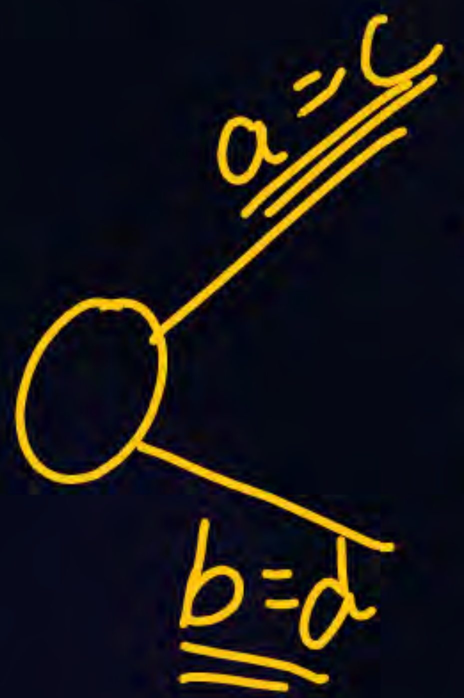
✗ CFL

$a^i b^j c^k \mid j = i + k$

$a^i b^{i+k} c^k$

$\underset{Pu}{a^i}\ \underset{Pop}{b^i}\ \underset{Pu}{b^k}\ \underset{Pop}{c^k}$ ✓DCFL

$a^i b^j c^k d^l \,/\, i=k \;\&\; j=l$



✓CFL

×DCFL

×Reg

$a^i b^j c^k d^l \,/\, i = k$ and $j = l$

XCFL

$a^n b^m c^n d^m$

PDA ✗ CFL ✗

CSL ✓

$a^m b^l c^k d^n \,/\, m, l, k, n \geq 1$

$a^+ b^+ c^+ d^+ \rightarrow$ Reg

$a^n b^{4m} / n, m \geq 1$

$\downarrow$

Reg

$\{a^3, a^8, a^{13}, a^{18} \ldots\ldots\}$

$\underline{\underline{aaa}}(aaaaa)^*$

RL

$a^{2n+1} / n \geq 1$

$\downarrow$

odd no of a's

$\downarrow$

Reg

$a^{n^n} / n \geq 1$

× Reg

× CFL

✓ CSL

$L = \{ w / w \in (a,b)^*,\ |w| \geq 100 \}$

$\downarrow$

Reg

$$L = \{ w / w \in \{a,b\}^* \text{ and } n_a(w) = n_b(w) \}$$

$$\downarrow$$

✓ DCFL

$$L = \{ w / w \in (a,b,c)^*, n_a(w) = n_b(w) = n_c(w) \}$$

✓ CSL ✗ $\underline{\underline{\text{C}}}$FL

$L = \{ \omega \mid \omega \in \{a, b\}^*, n_a(\omega) \geq n_b(\omega) + 1 \}$

✓ DCFL ✓

✓ CFL ✓

✓ CSL ✓

Elimination of ϵ-productions:-

$S \to AB$
$A \to aAA / \epsilon$
$B \to bBB / \epsilon$

$S \to AB / A / B / \epsilon$
$A \to aAA / aA / a$
$B \to bBB / bB / b$

Nullable (A, B, S)

when S is nullable, ϵ is in the language, we cannot eliminate all ϵ productions

There will be one ϵ production '$S \to \epsilon$'

$S \to A\,\underline{ba}\,C$

$A \to \underline{BC}$

$B \to b / \epsilon$

$C \to D / \epsilon$

$D \to d$

$S \to AbaC / baC / Aba / ba$

$A \to BC / C / B$

$B \to b$

$C \to D$

$D \to d$

✓

Nullable = $\{B, C, A\}$

Eliminate unit productions:-

$S \rightarrow A$

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow \underline{\underline{a}}$

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow \underline{\underline{a}}$

$S \to \underline{AB}$

$A \to a$

$B \to \cancel{C} / b / a$

$C \to \cancel{D}\ a$

$D \to \cancel{E}\ a$

$E \to a$

$B \to \cancel{C} \to D \to E \to a$

$C \to \cancel{D} \to E \to a$

$D \to \cancel{E} \to a$

Elimination of useless productions:-

Sometimes a vertex cannot generate any string then it is useless.

Sometimes a vertex cannot be reached, then it is useless.

$S \to \cancel{AB} / \underline{a}$

$A \to \cancel{BC} / \underline{b}$

~~$B \to \underline{a}B / \underline{C}$~~

~~$C \to \underline{a}C / \underline{B}$~~

Deriving = $\{\underline{a}, \underline{b}, S, A, \}$

$S \to a$

~~$A \to b$~~

$S \to a$ ✓

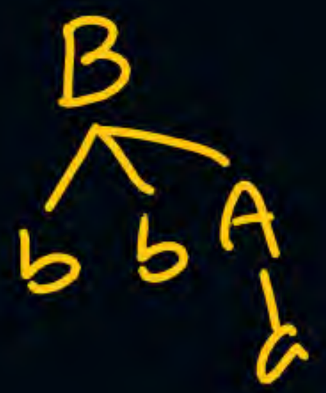
$S \to AB / \cancel{AC}$

$A \to aAb / bAa / a$

$B \to bbA / aaB / AB$

$\cancel{C \to abCA / aDb}$

$\cancel{D \to bD / aC}$



Deriving $= \{a, b, A, B, S\}$

$S \rightarrow \cancel{ABC} / BaB$

$A \rightarrow aA / \cancel{BaC} / aaa$

$B \rightarrow bBb / a$

$\cancel{C \rightarrow CA / AC}$

$S \rightarrow BaB$

$\cancel{A \rightarrow aA / aaa}$

$B \rightarrow bBb / a$

A is unreachable

Deriving = $\{a, b, B, A, S\}$

$S \rightarrow BaB$

$B \rightarrow bBb / a$

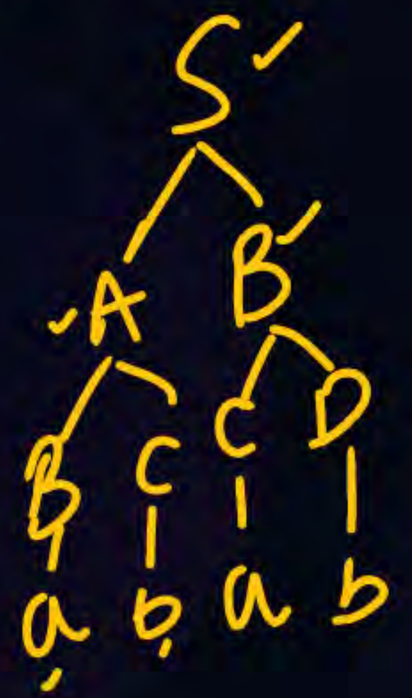
We can represent a grammar in many ways.

Two popular rep are CNF and GNF

Chomsky Normal form

CNF :- All the productions are of form $A \to BC$ or $A \to a$,
where A, B, C are vertices and 'a' is terminal

adv :-
1) length of each production is restricted
2) Derivation tree or parse tree is always binary tree



(iii) The no of steps required to derive a string of length $|w|$ is $\underline{\underline{(2|w|-1)}}$

$S \to AB$
$A \to a$
$B \to b$

$S \overset{1}{\Rightarrow} AB \overset{2}{\Rightarrow} aB \overset{3}{\Rightarrow} \underline{\underline{ab}}$

$|ab| = 2$ $\qquad 2(2)-1 = 3$ ✓

$|w| \qquad \underline{2|w|-1}$

(iv) It is easy to apply CYK algorithm

Convert the following grammar into CNF

$S \to aB / bA$

$A \to bAA / aS / a$

$B \to aBB / bS / b$

$S \to N_a B / N_b A$

$A \to \cancel{N_b AA} / N_a S / a / N_b X$

$B \to \cancel{N_a BB} / N_b S / b / N_b Y$

$X \to AA$, $Y \to BB$

$N_a \to a$

$N_b \to b$

CNF

GNF (Griebach Normal form)

→ If all the productions are of form $A \rightarrow a\alpha$ where $\alpha \in V^*$ then it is called GNF

$$\left.\begin{array}{l} A \rightarrow a \\ A \rightarrow aB \\ A \rightarrow aBC \\ A \rightarrow aBCD \end{array}\right\}$$

Adv:- of GNF:-

The no of steps required to generate a string of length $|w|$ is $|w|$

$$S \overset{①}{\Rightarrow} aBC \overset{②}{\Rightarrow} abC \overset{③}{\Rightarrow} \underline{abc}$$

3 length string

$\Downarrow$

3 steps

GNF is useful to convert CFG to PDA

$S \rightarrow aSb / ab$

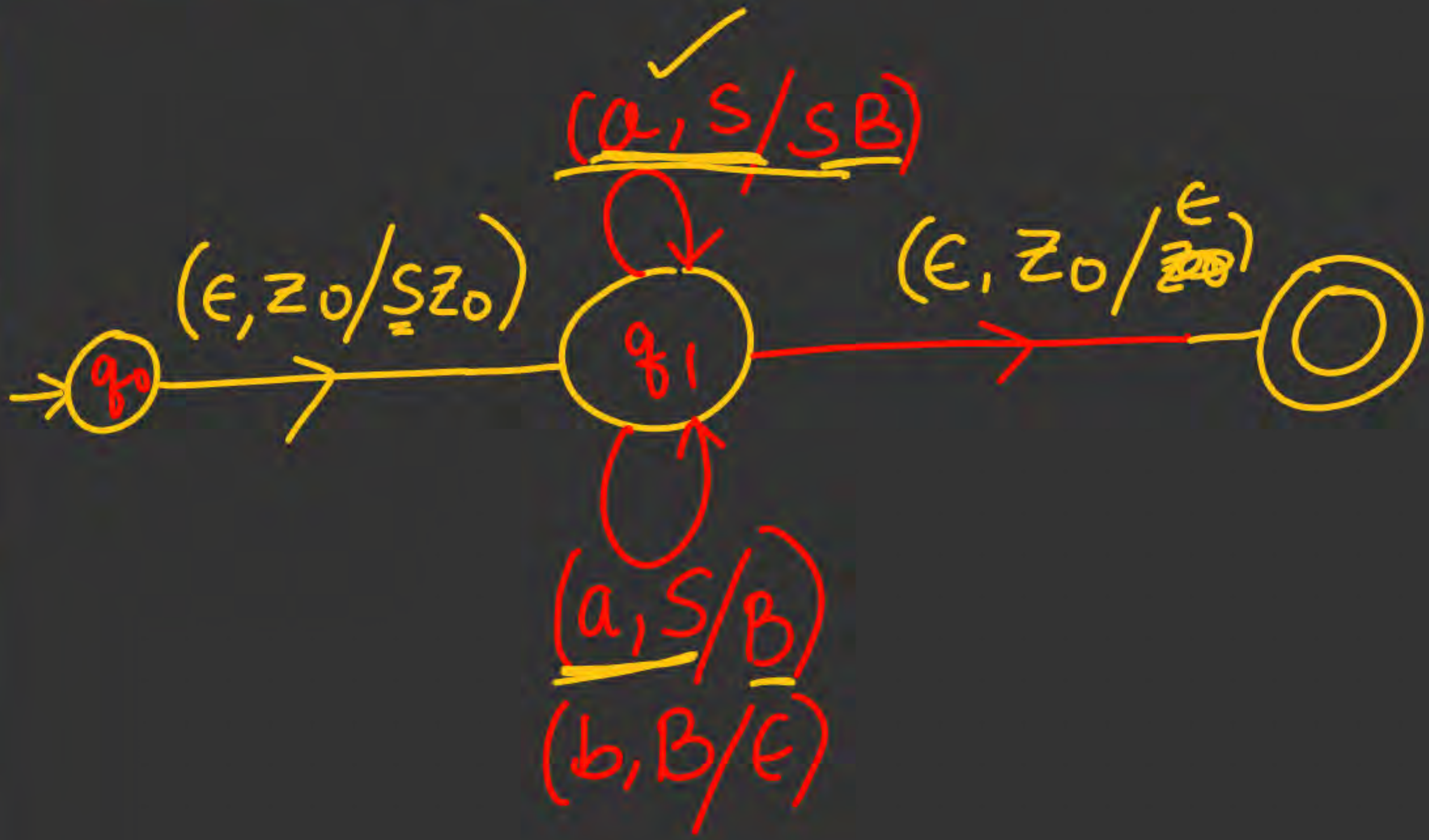
$\Downarrow$ GNF

$S \rightarrow aSB / aB$

$B \rightarrow b\epsilon$

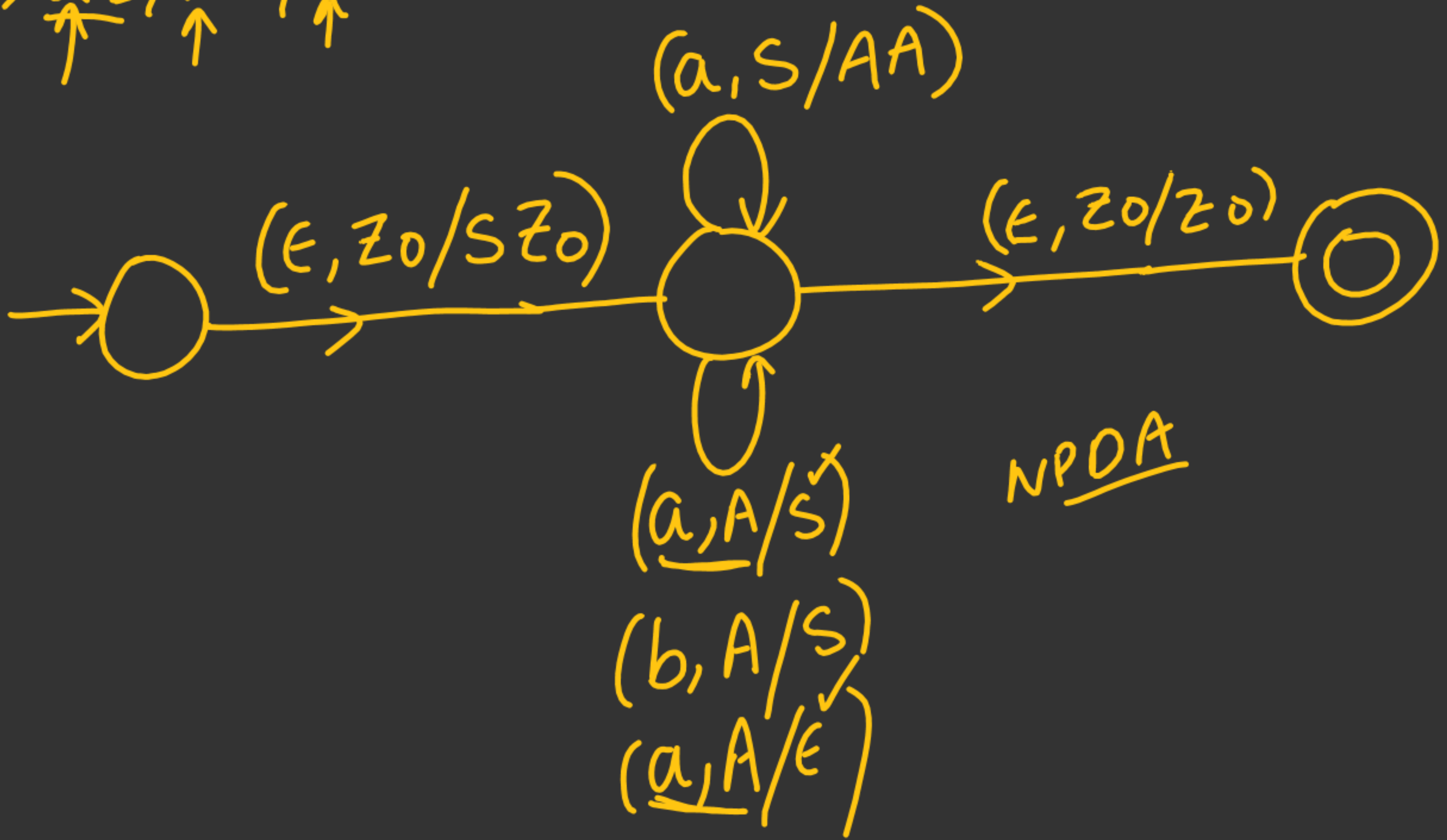
$\Downarrow$

PDA



$S \to aAA$

$A \to aS / bS / a\epsilon$



THANK - YOU